

Nunchi

The Agent Coordination Plane.

A durable runtime for production agents,
with verifiable coordination across organizations.

The model is commoditizing.

The scaffold is the moat.

T W O C O M P O N E N T S , O N E A R C H I T E C T U R E

Roko *the cognitive runtime*

Open-source Rust runtime. Plan, route, gate, persist,
replay. Functional and self-hosting today.

Korai *the verifiable substrate*

Agent-native EVM L1. ERC-8004 native. Where local
memory becomes shared memory across organizations.

177K LOC Rust · 18 crates · Apache 2.0

WHAT THIS DOCUMENT ARGUES

Models execute. Nunchi coordinates.

The model is commoditizing. The scaffold is the moat.

Frontier capability per dollar trends down. Switching cost between model providers approaches zero. The variable that separates a production agent from a prototype is no longer the model — it is the system around the model: identity, memory, calibration, policy, settlement, audit. We call that system the Agent Coordination Plane. We are building it.

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- 05 The category
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Every layer of the agent stack is funded. Except the one between them.

Three signals from the last ninety days.

FUNDING

\$750M

Google Cloud agentic partner fund

April 22, 2026

Hyperscaler-led capital flowing to ecosystem deployment, not into the coordination layer itself.

DEMAND

1,445%

Surge in enterprise multi-agent inquiries

Q1 2024 to Q2 2025

Gartner. Buyer intent is no longer the constraint. The constraint is making fleets of agents work in production.

ADJACENT

\$60M

Orkes Series B for single-org orchestration

April 23, 2026

Adjacent layer is funded. Cross-org coordination — the layer above — is empty.

*The empty layer between application and execution is the category.
It does not yet have a winner.*

02
T H E C L O C K

EU AI Act Article 50 enforces August 2, 2026.

Compliance buyers move when the date is on the calendar.



• R E A D I N E S S •

26.2%

of EU enterprises

have begun concrete Article 50 compliance activity (Deloitte, Q1 2026). 73.8% have not.

• •

• P R I C I N G •

20–30%

vendor price uplift

already visible on AI procurement contracts that include compliance attestation (Raconteur, April 2026).

• •

• P E N A L T Y •

\$15M

or 3% of turnover

maximum penalty under Article 99 for Article 50 transparency failures. The audit trail is the asset.

• •

Vanta turned SOC 2 into \$100M ARR. OneTrust turned GDPR into \$5B+.
Article 50 is the next compliance category. The audit trail is native, not bolted on.

AI replacing humans is reversing — for system reasons, not capability reasons.

Three reversals. Twenty months. Pattern accelerating.

• M A Y 2 0 2 5 •

Klarna

Reversed AI-only customer service.

5,500 then 3,400

headcount cut, then reversed

"Cost unfortunately seems to have been a too predominant evaluation factor — what you end up with is lower quality."

— Sebastian Siemiatkowski, CEO

• A U G 2 0 2 5 •

Commonwealth Bank of Australia

Rehired 45 customer-service staff after voice bot failure.

rising

call volumes that the bot was meant to reduce

Union demonstrated bot-handled call volumes were rising, not falling. Customers escalated to humans by default.

— Finance Sector Union evidence

• J A N 2 0 2 6 •

NYC MyCity chatbot

Mamdani administration killed the program.

\$500K

annual cost of a 'functionally unusable' bot

Killed in Q1 2026. Cited cost and unreliability — not modelling capability.

— Office of the Mayor

*These didn't fail at modeling. They failed at coordination, audit,
and accountability — the layer no model provider sells.*

The model is not the variable. The system is.

Three findings from the last twelve months of agent research.

41–86%

Multi-agent failure rates across 1,642 production traces.

MAST taxonomy, Cemri et al. (Berkeley, NeurIPS 2025) — 14 distinct failure modes, three classes: system design, inter-agent misalignment, task verification. Inter-annotator agreement $\kappa = 0.88$.

64%

Tasks where a single well-tooled agent matches multi-agent ensembles.

Princeton NLP. Adding agents adds failure surface, not capability. The marginal agent costs more than it earns.

~45%

Single-agent accuracy threshold beyond which more agents hurt.

Google DeepMind, Towards a Science of Scaling Agent Systems (arXiv 2512.08296, v3 April 2026). Coordination errors compound geometrically — naive multi-agent setups show 17.2× error amplification.

Adding agents doesn't add capability.

It adds failure modes that no model can solve.

What's missing isn't a better model. It's the plane that decides above them.

A fleet of production agents is a distributed system with non-human actors. Models execute. Frameworks describe. But neither answers the questions a production deployment actually faces:

- *Which agent should act?*
- *With which memory?*
- *Calling which model, at what cost?*
- *Under whose policy?*
- *With what proof for the regulator?*

These are coordination questions, not capability questions. They are the same questions every distributed system has had to answer since networks acquired control planes in 2007.

ARCHITECTURAL PRECEDENT

Ethane

Stanford, 2007 (Casado et al.)

Networks separated the data plane (forwarding) from the control plane (decisions about what to forward). The control plane became Software-Defined Networking. Nicira commercialized it. VMware acquired it for \$1.26B.

the pattern repeats

Production agents need the same separation. Frameworks describe what an agent does. Models execute. The plane in the middle decides which agent, which memory, which model, at what cost, under which policy.

That plane is Nunchi.

WHERE NUNCHI SITS

The plane between the planes.

APPLICATION LAYER

Frameworks. Agents. Workflow logic.

LangChain · CrewAI · Mastra · AutoGen · Orkes · AGENTS.md

AGENT COORDINATION PLANE

Nunchi — identity, routing, knowledge, policy, settlement
Roko (runtime) + Korai (verifiable substrate)

EXECUTION LAYER

Models. Tools. Durable execution. Identity. Payment rails.

Temporal · Keycard · x402 · ERC-8004 · MCP · A2A · Model APIs

*Ethane separated networking's data plane from its control plane in 2007.
Nunchi does the same for production agents.*

Two components. One architecture.

Roko is the runtime every agent passes through. Korai is where their lessons compound.

• R O K O · R U N T I M E •

Roko

the cognitive runtime

Open-source Rust runtime that turns a model call into a governed event. Predict cost. Route to the cheapest qualified model. Gate before commit. Persist usefully. Replay deterministically.

W H A T I T S H I P S W I T H

- **CascadeRouter** — cost-aware model routing across Haiku to Opus.
- **Gate pipeline** — 11 gates, 7 rungs — early-exit on test pass.
- **EpisodeLogger** — deterministic replay across non-deterministic LLMs.
- **NeuroStore** — HDC-indexed knowledge that persists across runs.
- **Control plane** — policy, identity, audit bundle per execution.

Open-source. Functional. Self-hosting.

177K LOC Rust · 18 crates · Apache 2.0

• K O R A I · S U B S T R A T E •

Korai

the verifiable substrate

An agent-native EVM L1 where Roko episodes settle. Where local memory becomes shared memory. Where reputation is an attestation, not a marketing claim. Where coordination across organizational trust boundaries becomes verifiable.

W H A T I T M A K E S P O S S I B L E

- **ERC-8004** — identity / reputation / validation registries.
- **HDC fabric** — shared semantic substrate, not just tool calls.
- **ZK behavior** — prove behavioral fingerprints without revealing them.
- **Cooperative** — clearing — match, route, settle, score with proofs.
- **Settlement** — agent obligations clear like financial obligations.

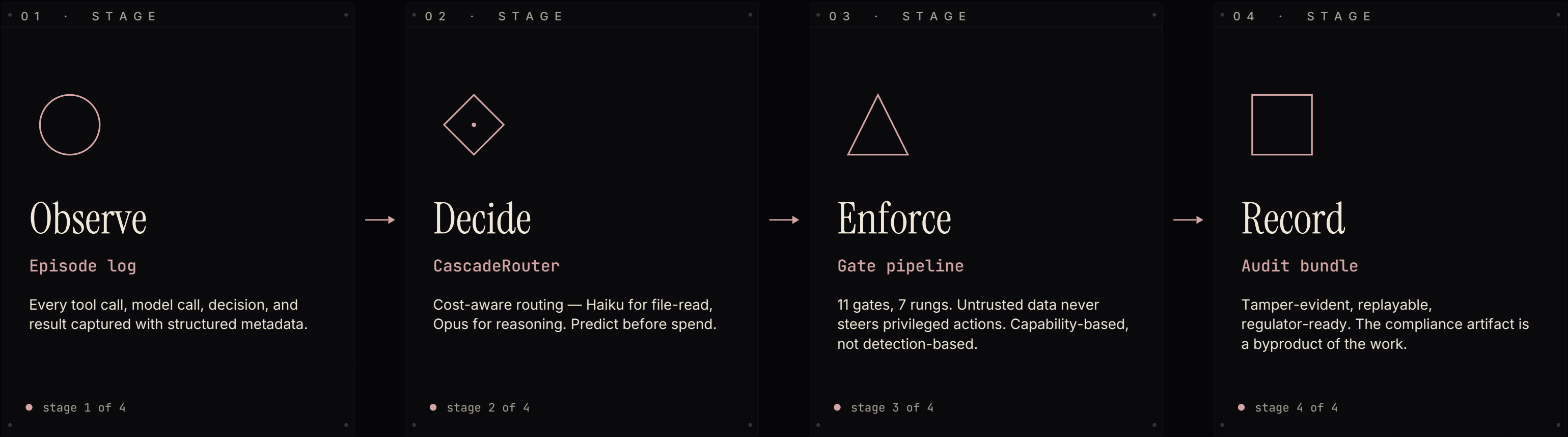
Specified. Sequenced after runtime adoption.

ERC-8004 native at genesis · optional verification layer

Roko is what every agent run passes through. Korai is what makes the runs compound across organizations.

How it closes the loop.

Four functions, four primitives, one inspectable trace per execution.



Every agent run produces an inspectable trace.

Replay isn't a transcript. It's a verifiable reconstruction.

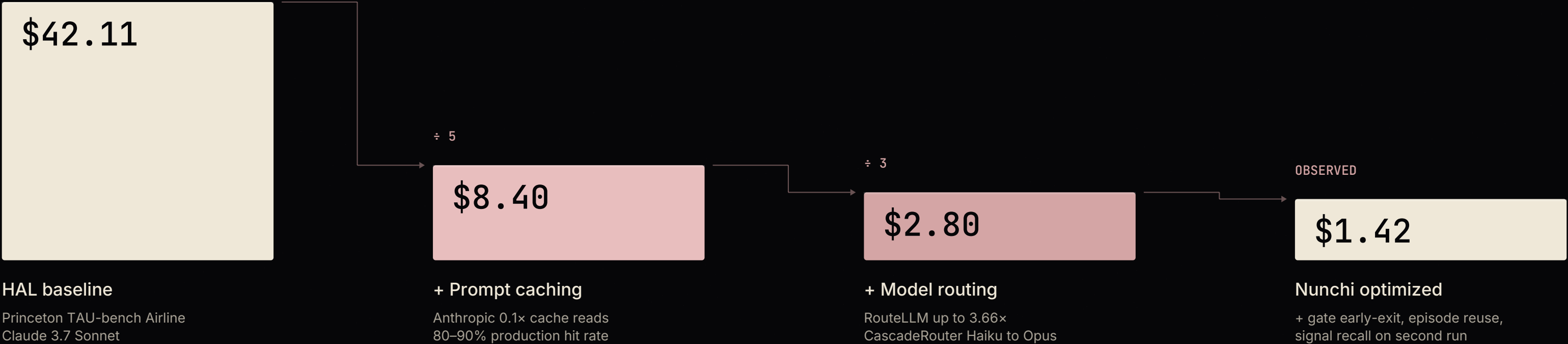
→ regulator-ready, compliance-native, audit-portable

08
THE WEDGE

Same task. Order of magnitude cheaper. With a receipt.

Compounded primitives. Each decomposable. Each with a citation.

PUBLISHED



30× compound cost reduction
\$42.11 reduced to \$1.42 per task

FOOTNOTE · METHODOLOGY

HAL costs exclude prompt caching by design. Anthropic publishes 0.1× cache-read pricing — applying caching on top of HAL is mathematically valid, and is footnoted in the published Nunchi methodology along with task IDs, model versions, cache hit rate, and gate exit rate. Reproducible.

Cost is the wedge. Knowledge is the moat.

Every agent run leaves a deposit. The deposit is the moat.

• KNOWLEDGE •



Knowledge

HDC-indexed Signals

Content-addressed memory objects. Cheap similarity, deterministic recall, decay-aware. Available to every future agent.

30K tokens / hit

saved when a Signal recalls

• CALIBRATION •



Calibration

Predict-publish-correct

Every agent registers a falsifiable prediction before acting. The residual updates priors. Cost and risk forecasts get sharper over time.

1.78×

compute reduction at iso-accuracy

• REPUTATION •



Reputation

ERC-8004 attestation

Behavioral history is on-chain, attestable, portable across deployments. An agent's track record is not lost when its container dies.

22,900

registrations in first 3 days

• SETTLEMENT •



Settlement

Cooperative clearing

Match, route, settle, score with verifiable optimality certificates. Agent obligations clear like financial obligations.

\$50M

x402 cumulative volume

The thousandth agent joins smarter than the first.

A fork starts empty. Code can be copied. Compounded coordination cannot.

Every adjacent layer is funded. The coordination plane is not.

Five adjacent companies. Five funded layers. One empty cell.

C O M P A N Y	R O U N D	V A L U A T I O N	L A Y E R	S C O P E
Temporal	Series D · Feb 2026	\$5B	Durable execution	single-org
LangChain	Series B · Oct 2025	\$1.25B	Agent framework	developer
Braintrust	Series B · Feb 2026	~\$800M	AI evaluation	single-org
Orkes	Series B · Apr 2026	~\$300M	Workflow orchestration	single-org
Keycard	Series A · Oct 2025	~\$200M	Agent identity / auth	single-org
Nunchi	Series A · target	[empty]	Coordination plane	cross-org

The empty cell is the pitch.

Five questions. Five owners. Only one of them is coordination.

	Nunchi	Temporal	Orkes	Keycard	LangChain
Cross-org coordination		×	×	×	×
Cost-aware model routing		×	×	×	×
Verifiable behavioral identity		×	×	partial	×
Shared knowledge substrate		×	×	×	×
Open-source runtime		×		×	

Temporal owns 'did this code run.' Orkes owns 'did this workflow complete.'
Keycard owns 'is this agent authorized.' LangChain owns 'what should the agent do step by step.'
Nunchi owns: did the right agent, with the right memory, at the right price, produce a verifiable result?

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THE RAILS ARE CRYSTALLIZING

Each protocol owns a rail. Coordination across them is empty.

MCP, A2A, ERC-8004, x402 — donated, standardized, deployed in the last 18 months.

• M C P •

MCP

Model Context Protocol

97M

monthly SDK downloads

10K+

active public servers

OWNS

Tool & context access

Anthropic / AAIF

Donated to Linux Foundation, Dec 2025

• A 2 A •

A2A

Agent-to-Agent Protocol

150+

organizations in production

v1.2

stabilized April 2026

OWNS

Agent interoperability

Google / Linux Foundation

First-year mark April 9, 2026

• E R C - 8 0 0 4 •

ERC-8004

Trustless Agents

22.9K

registrations in 3 days

March 17

Ethereum mainnet, 2026

OWNS

Identity · reputation · validation

Ethereum Foundation

Phala, Oasis, TRON, Morph backing

• X 4 0 2 •

x402

HTTP Payment Protocol

\$50M

cumulative volume

165M+

transactions

OWNS

Machine-native payments

Coinbase + Cloudflare

AWS · Stripe · Visa · Mastercard back

Each protocol solves one rail.

The plane that decides which agent uses which rail under whose policy is what Nunchi owns.

Three flywheels. Each turn deepens the next.

• K N O W L E D G E •

01

Knowledge

Every run deposits memory. Memory compounds across runs.

Roko persists every successful pattern as an HDC-indexed Signal. Korai makes those Signals available across organizations through ERC-8004 attestation. The thousandth agent pulls 30,000 tokens of context the first agent never had.

30K

tokens recalled per Signal hit

Code can be forked.
Lived coordination history cannot.

• P R O T O C O L •

02

Protocol

Deep integration creates architectural switching cost.

MCP is 97M monthly downloads. A2A is 150 organizations. ERC-8004 ships natively. x402 is 165M transactions and AWS / Stripe / Visa / Mastercard backing. Deep integration with all four creates switching costs that are architectural, not contractual.

Q4 2026

window before payment rails lock identity

The protocols are crystallizing.
The plane above them is not.

• R E G U L A T O R Y •

03

Regulatory

August 2, 2026 is a known spending trigger.

Article 50 transparency obligations enforce on a calendar date. Vanta turned SOC 2 into \$100M ARR. OneTrust turned GDPR into a \$5B+ company. Nunchi makes Article 50 audit trails native — the compliance record is a byproduct of the work, not a separate spend.

\$15M

max penalty under Article 99

The audit bundle never migrates.
The customer never leaves.

Precedent — Vanta: \$100M ARR from SOC 2 automation. OneTrust: \$5B+ from GDPR tooling. Compliance is a recurring revenue category.

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M A R K E T

Anchor: Non-Human Identity governance.

Named Gartner category. Premium acquisition multiples already confirmed.

• P R I M A R Y A N C H O R •

\$10.71B

→

\$25.65B

NHI Access Management market, 2025 to 2033 · 11.53% CAGR

80:1+

machine-to-human identity ratio (Gartner, 2026)

\$400M+

raised in NHI governance category, 2025

\$25B

Palo Alto / CyberArk acquisition (Feb 2026)

Source: Frost & Sullivan / Gartner. Premium multiples — CyberArk acquired at 11× revenue.

• S E C O N D A R Y •

AI Agents market, 2025 to 2030

\$7.84B

→

\$52B

46–50% CAGR

1,445%

72%

surge in enterprise multi-agent inquiries

Global 2000 running agents beyond pilot

Anchor on the named, verified, premium-multiple category.

Use the agent market as the larger context, not the pitch.

The Agent Coordination Plane · Series A whitepaper · 2026

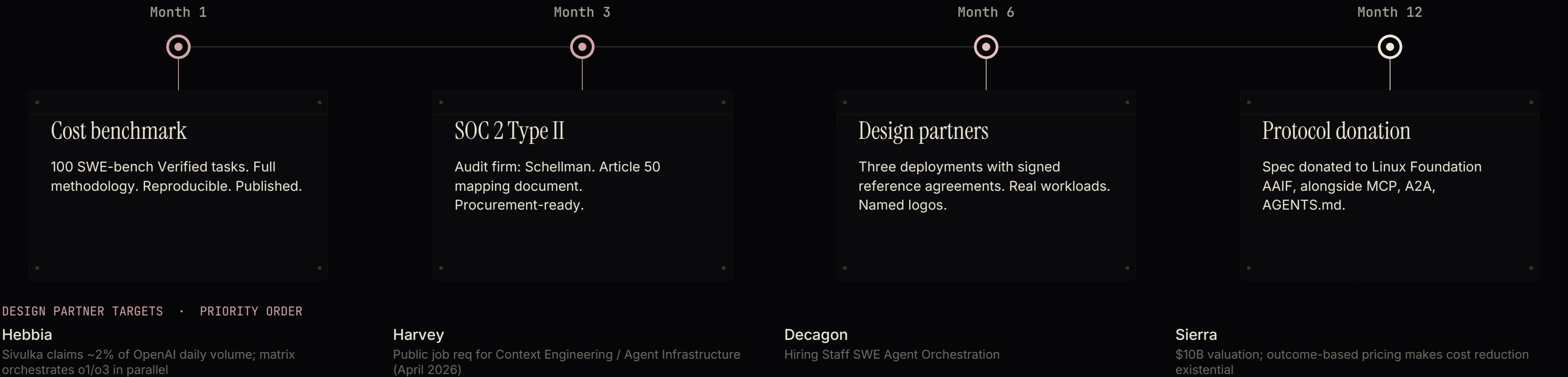
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WHAT THIS ROUND BUYS

Twelve months. Four milestones. Each unlocks the next.

Discipline of sequencing: developer adoption · benchmark · design partners · protocol surface.



Five lines.

The whole argument.

- 01 Production agents fail at coordination, not capability — the failure rate is 41–86% and the cause is system-level, not model-level.
- 02 Roko is the runtime wedge — it makes production agents cheaper, safer, replayable. Open source. Self-hosting.
- 03 Korai is the substrate where local traces compound — into shared memory, attestable reputation, verifiable settlement across organizations.
- 04 Every adjacent layer is funded — Temporal \$5B, LangChain \$1.25B, Orkes \$300M, Keycard \$200M. The coordination plane is empty. That's the category.
- 05 Models commoditize. Scaffolds compound. The model is the same. The system is the variable.

Models execute. Nunchi coordinates.

The model is commoditizing. The scaffold is the moat.